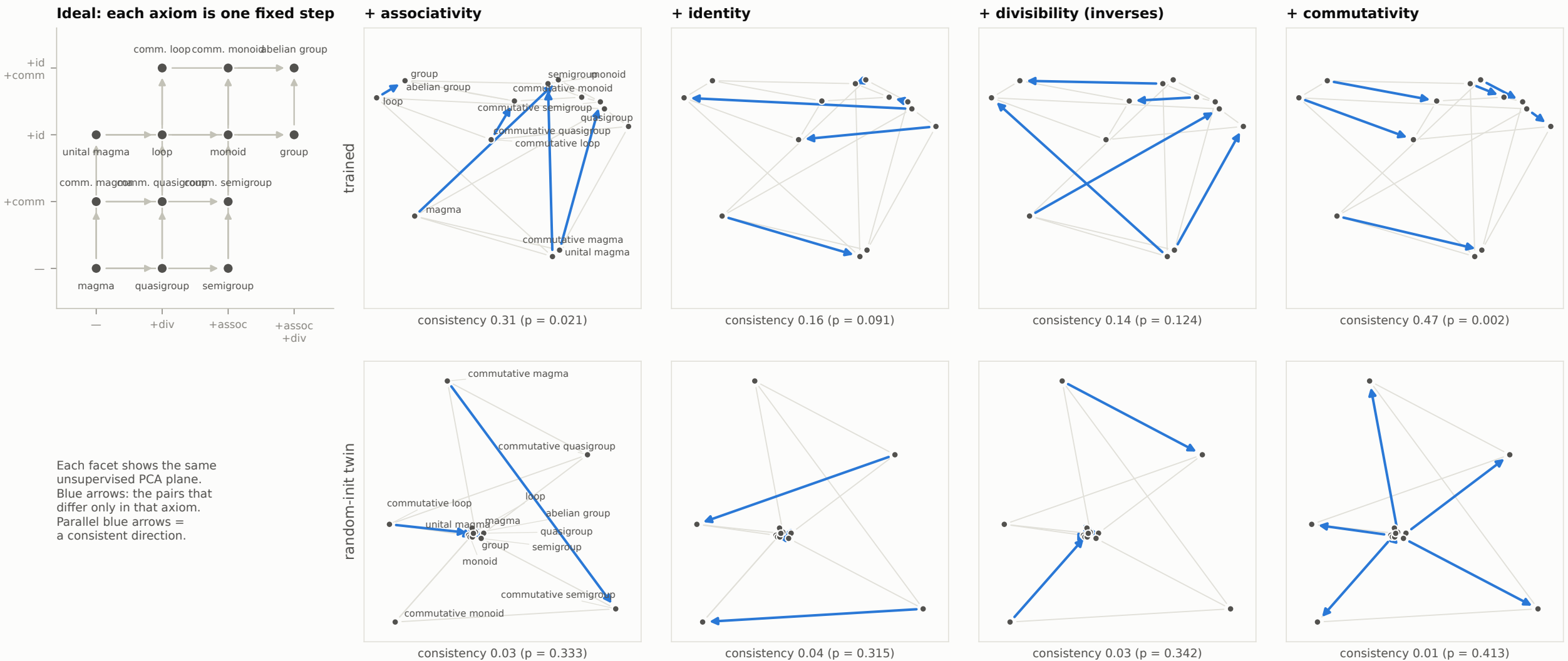
Ideal: each axiom is one fixed step
+ associativity
+ identity
+ divisibility (inverses)
+ commutativity
comm. loop
comm. monoid
abelian group
+id +comm
+id
unital magma
loop
monoid
group
comm. magma
comm. quasigroup
comm. semigroup
+comm
magma
quasigroup
semigroup
—
+div
+assoc
+assoc +div
trained
group
abelian group
loop
semigroup
monoid
commutative monoid
commutative semigroup
quasigroup
commutative quasigroup
commutative loop
magma
commutative magma
unital magma
consistency 0.31 (p = 0.021)
consistency 0.16 (p = 0.091)
consistency 0.14 (p = 0.124)
consistency 0.47 (p = 0.002)
random-init twin
commutative magma
commutative quasigroup
commutative loop
loop
abelian group
unital magma
magma
quasigroup
group
semigroup
monoid
commutative semigroup
commutative monoid
consistency 0.03 (p = 0.333)
consistency 0.04 (p = 0.315)
consistency 0.03 (p = 0.342)
consistency 0.01 (p = 0.413)
Each facet shows the same unsupervised PCA plane. Blue arrows: the pairs that differ only in that axiom. Parallel blue arrows = a consistent direction.